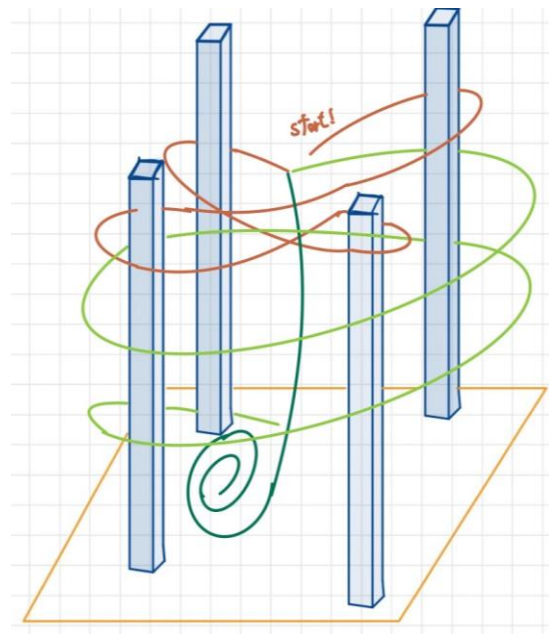
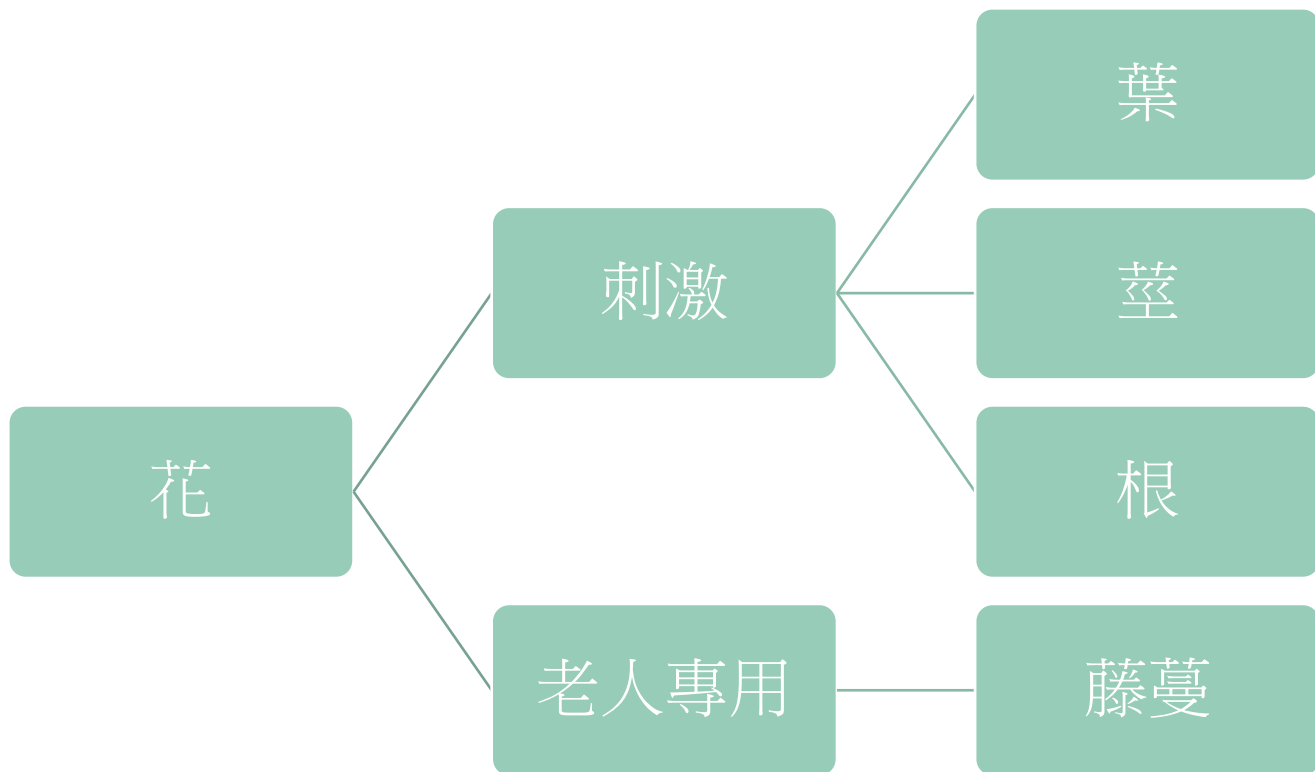


FINAL PRESENTATION

土實 1BT6

6 薛 羽 庭
3 莊 承 昌
5 廖 致 翔
9 楊 子 毅
4 陳 識 凱

✈️ track design



以植物貫穿整個軌道的結構

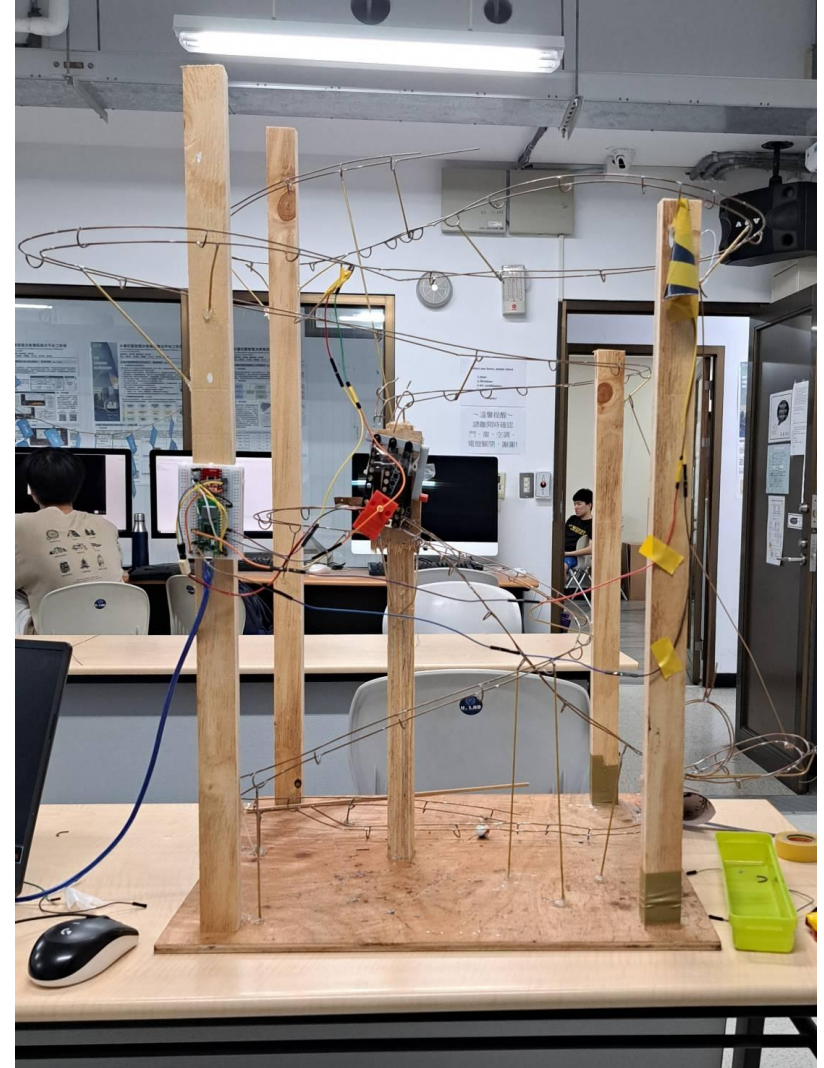
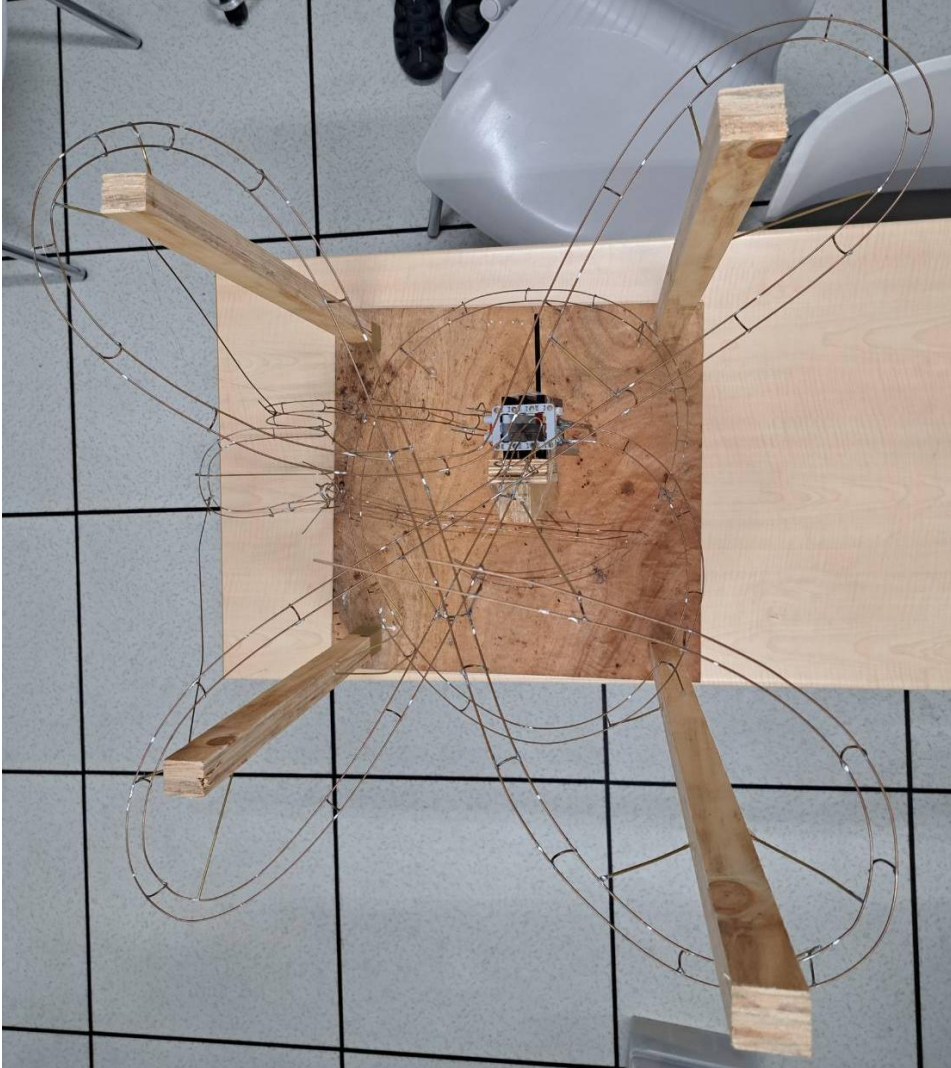
先以一朵花開場

接著一路從葉子，到莖，再到根

最後藤蔓一圈一圈地圍繞在植物周圍

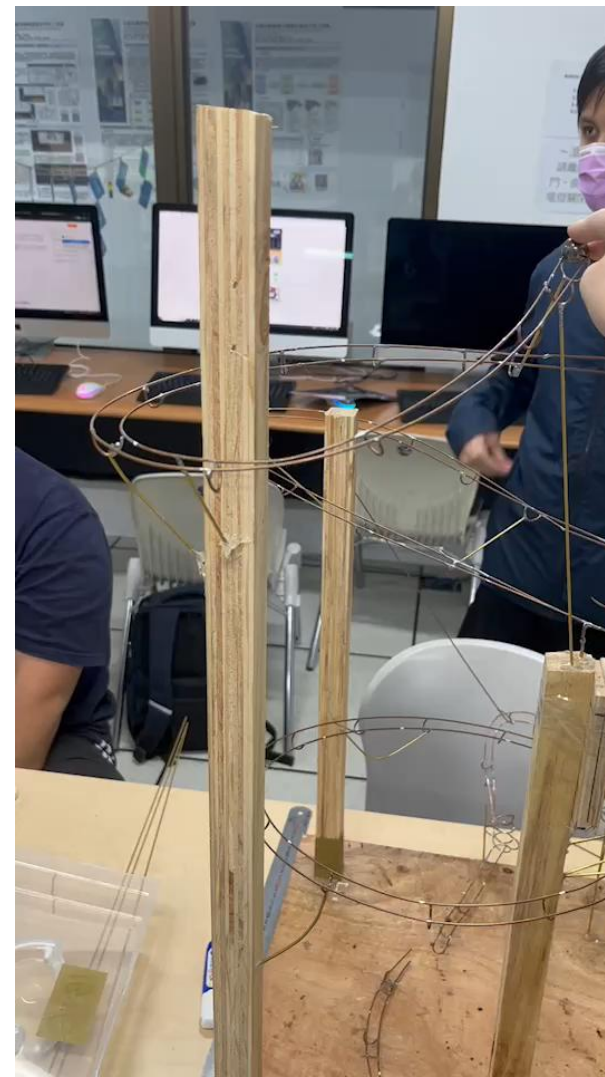
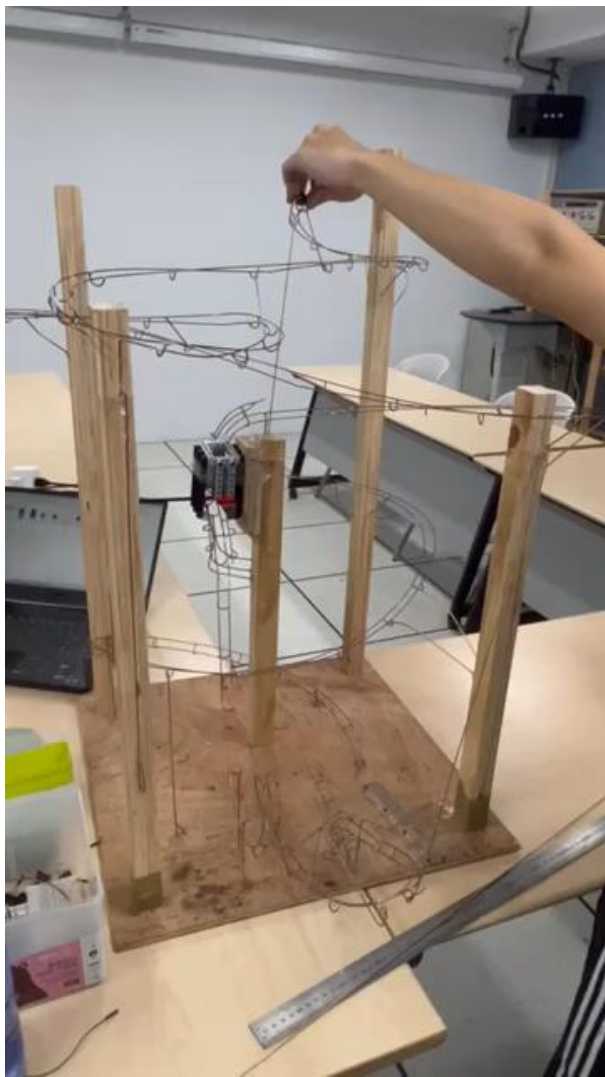
展現我們對大自然的熱愛

track design





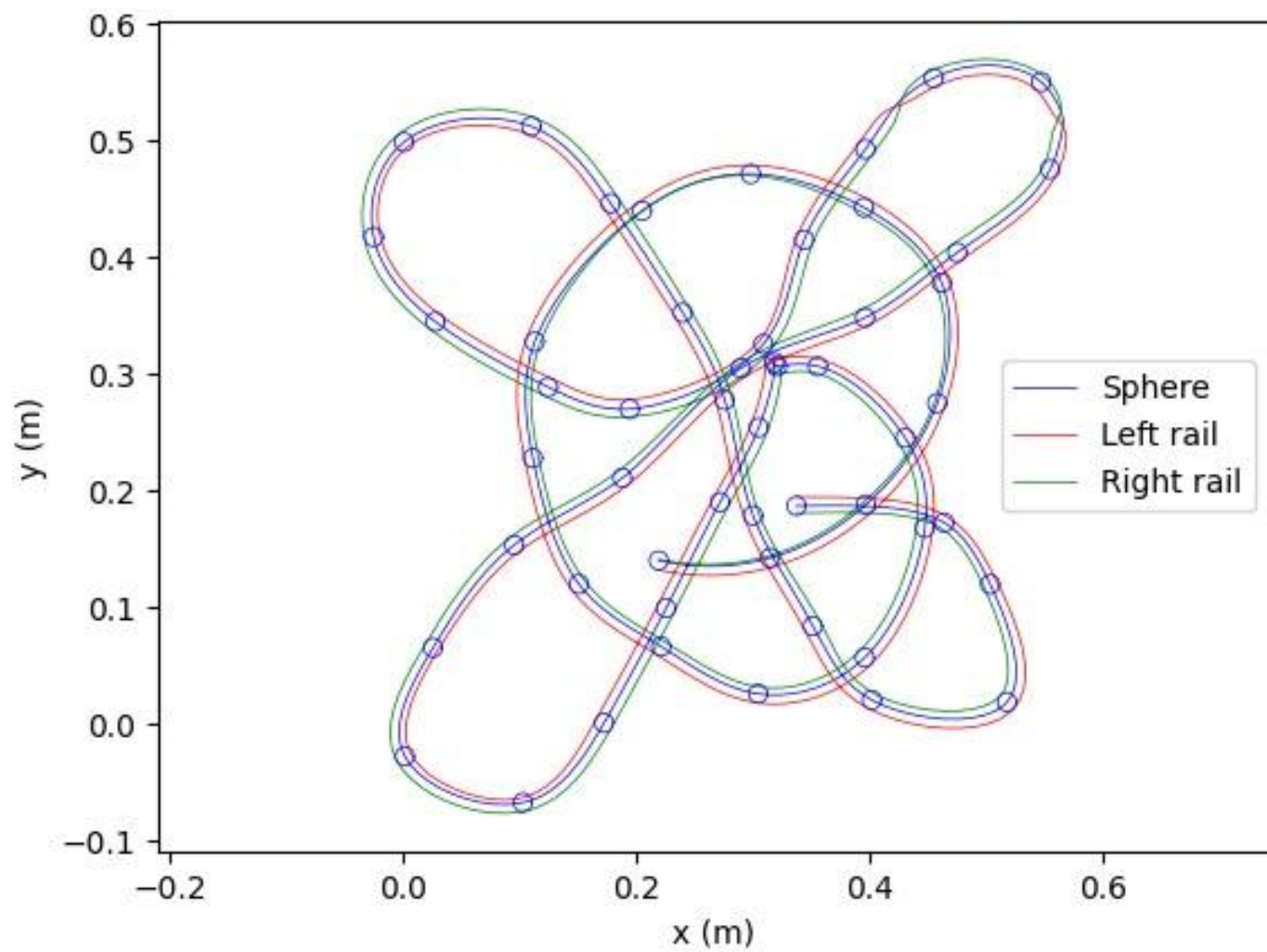
video



 track geometry

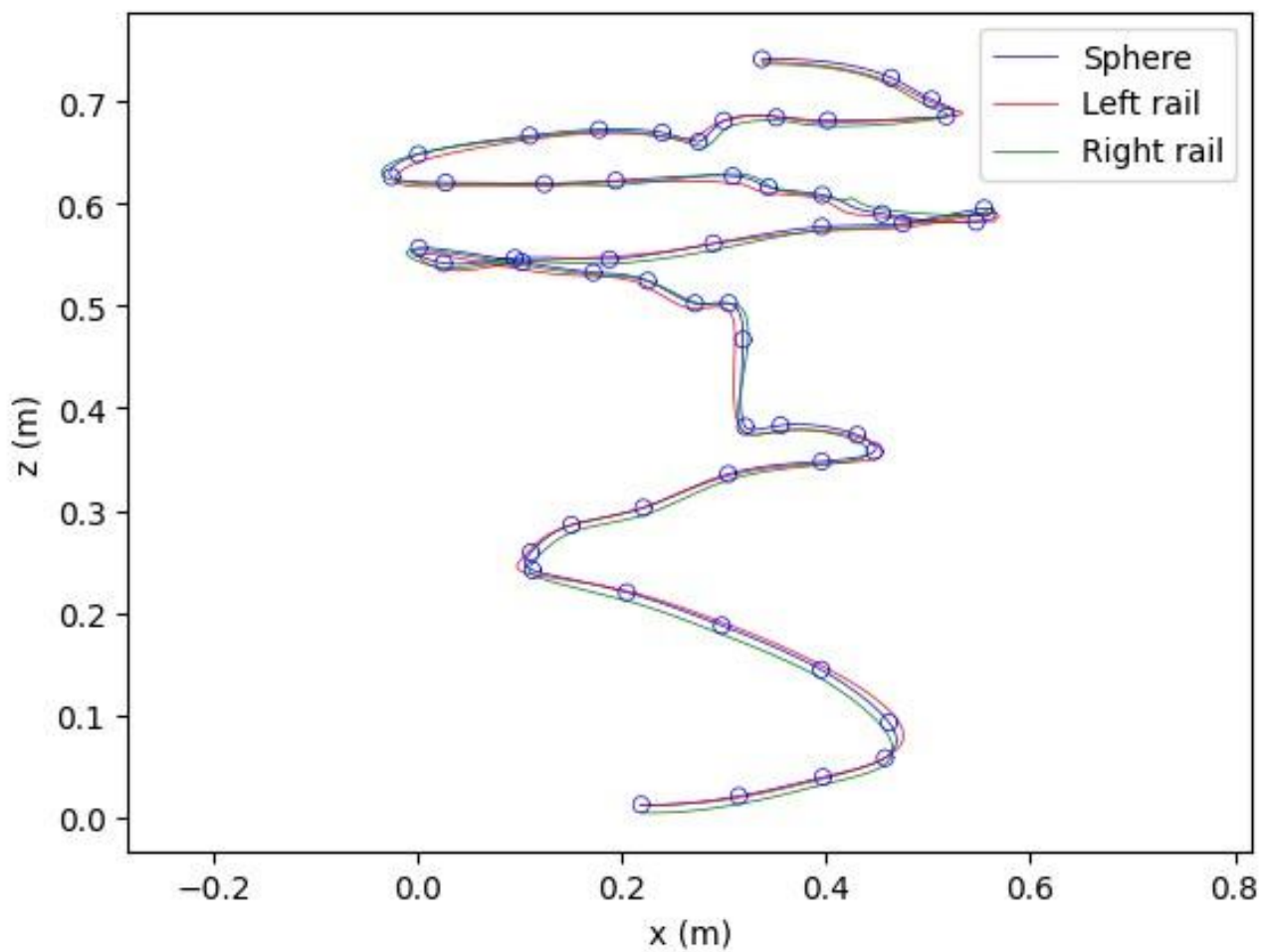


XY humdrum



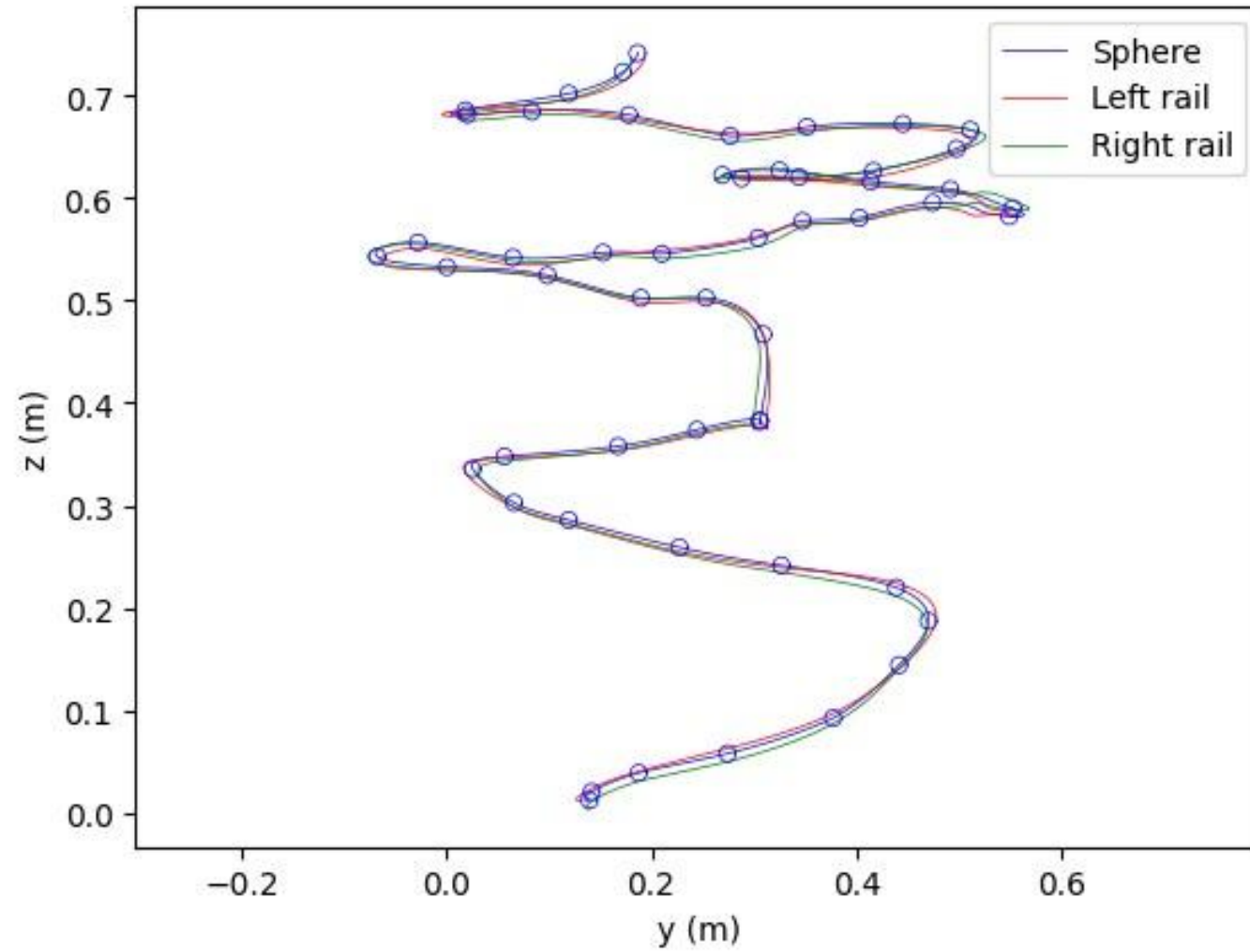


XZ humrum



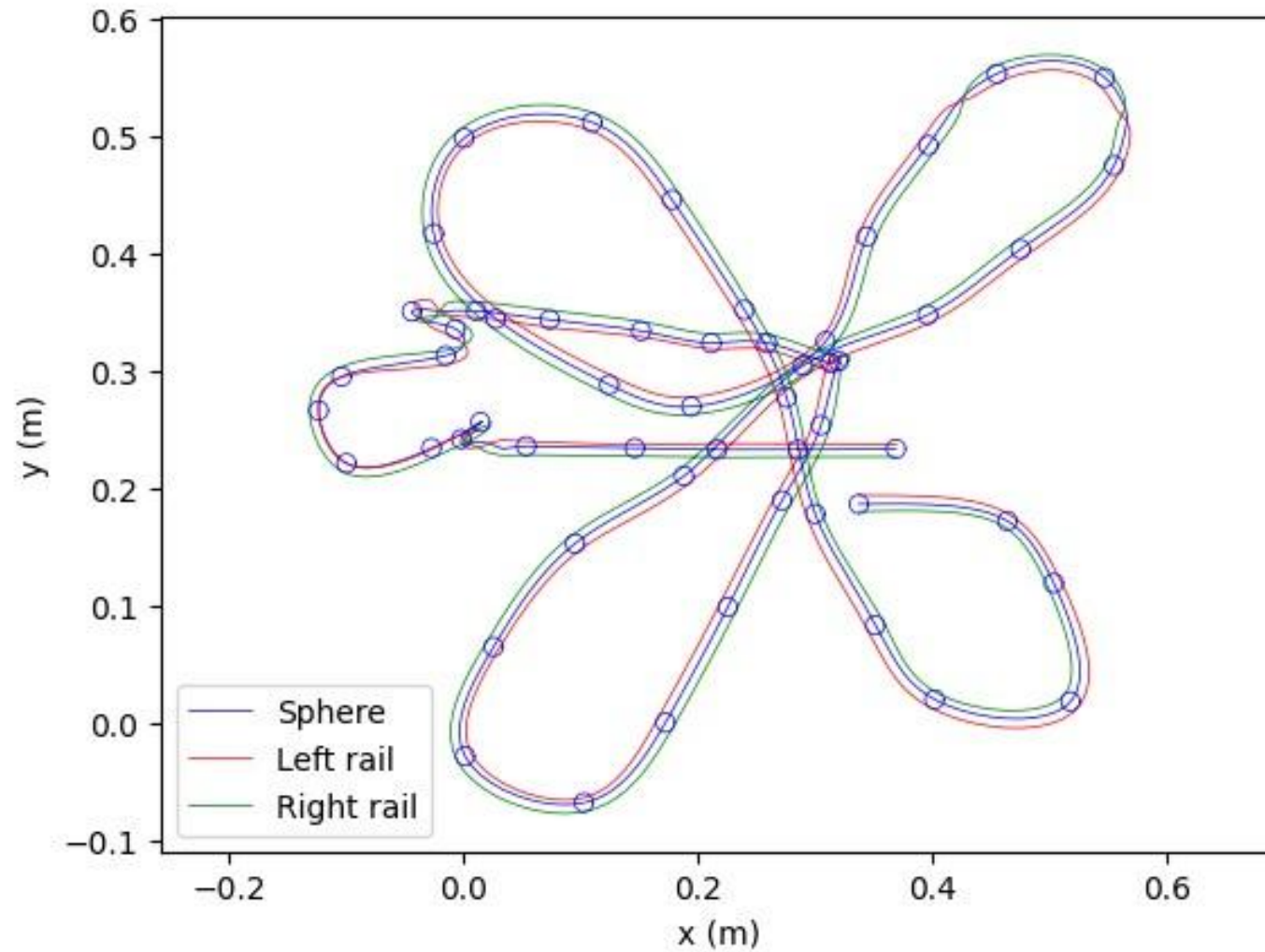


YZ humdrum



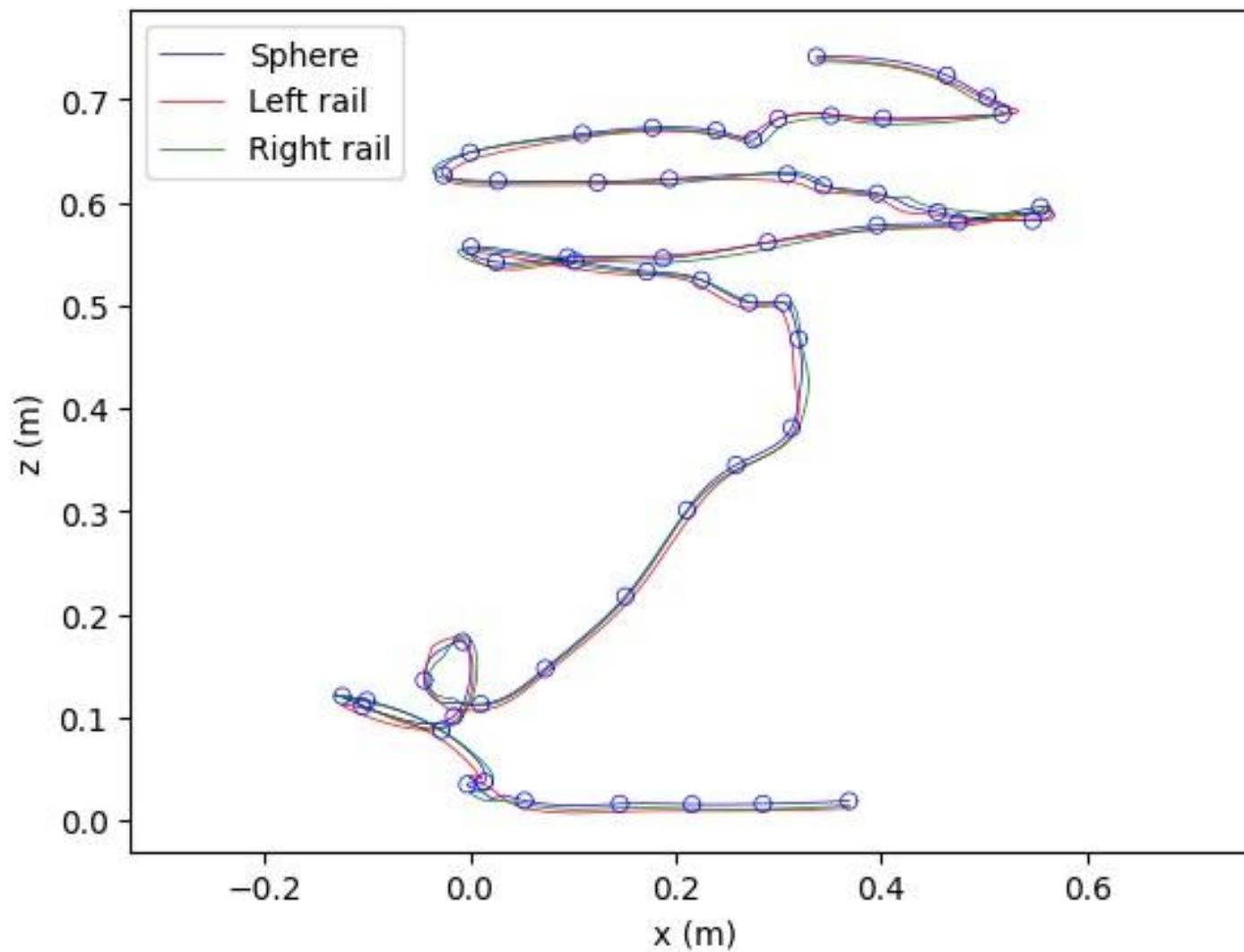


XY thrilling



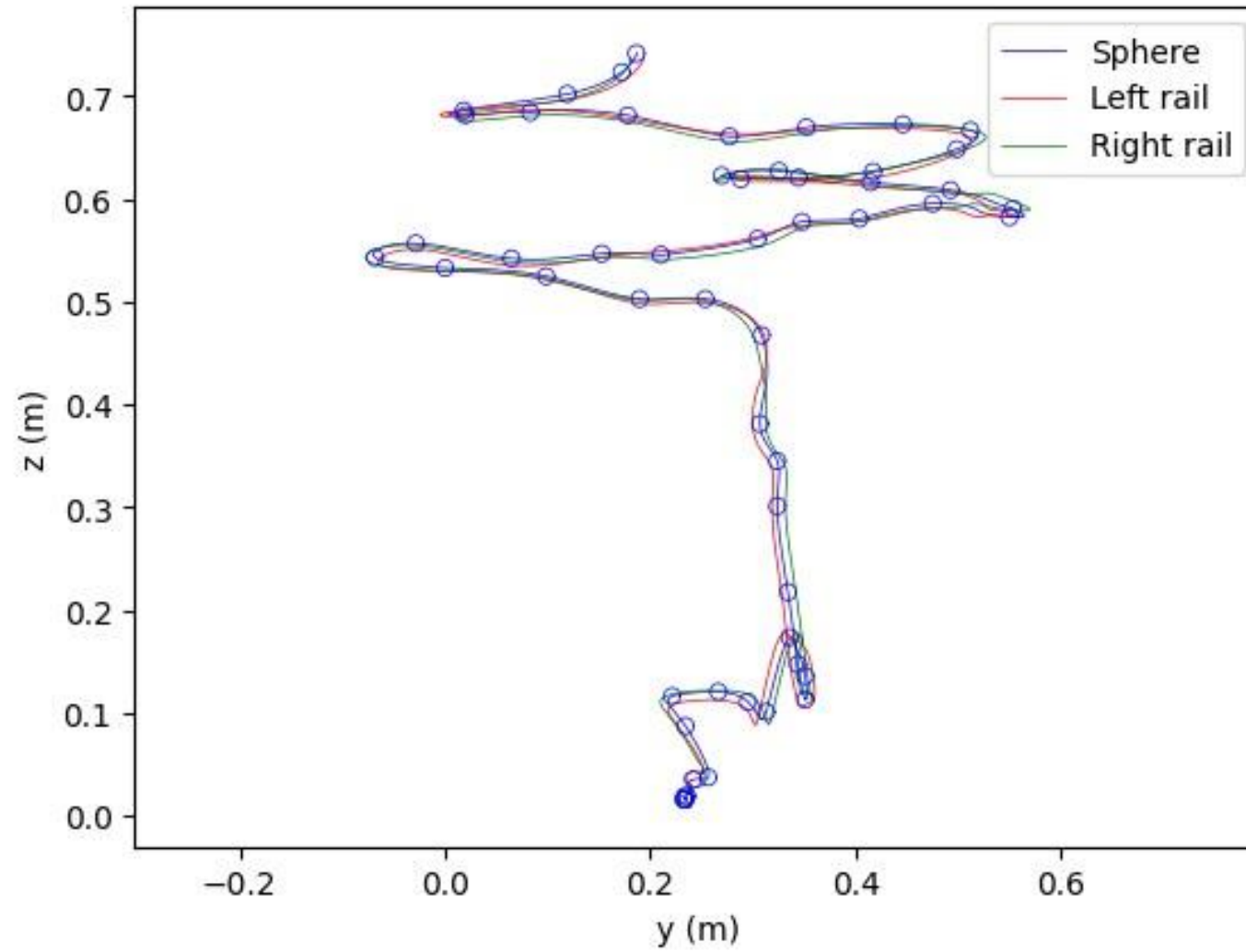


XZ thrilling



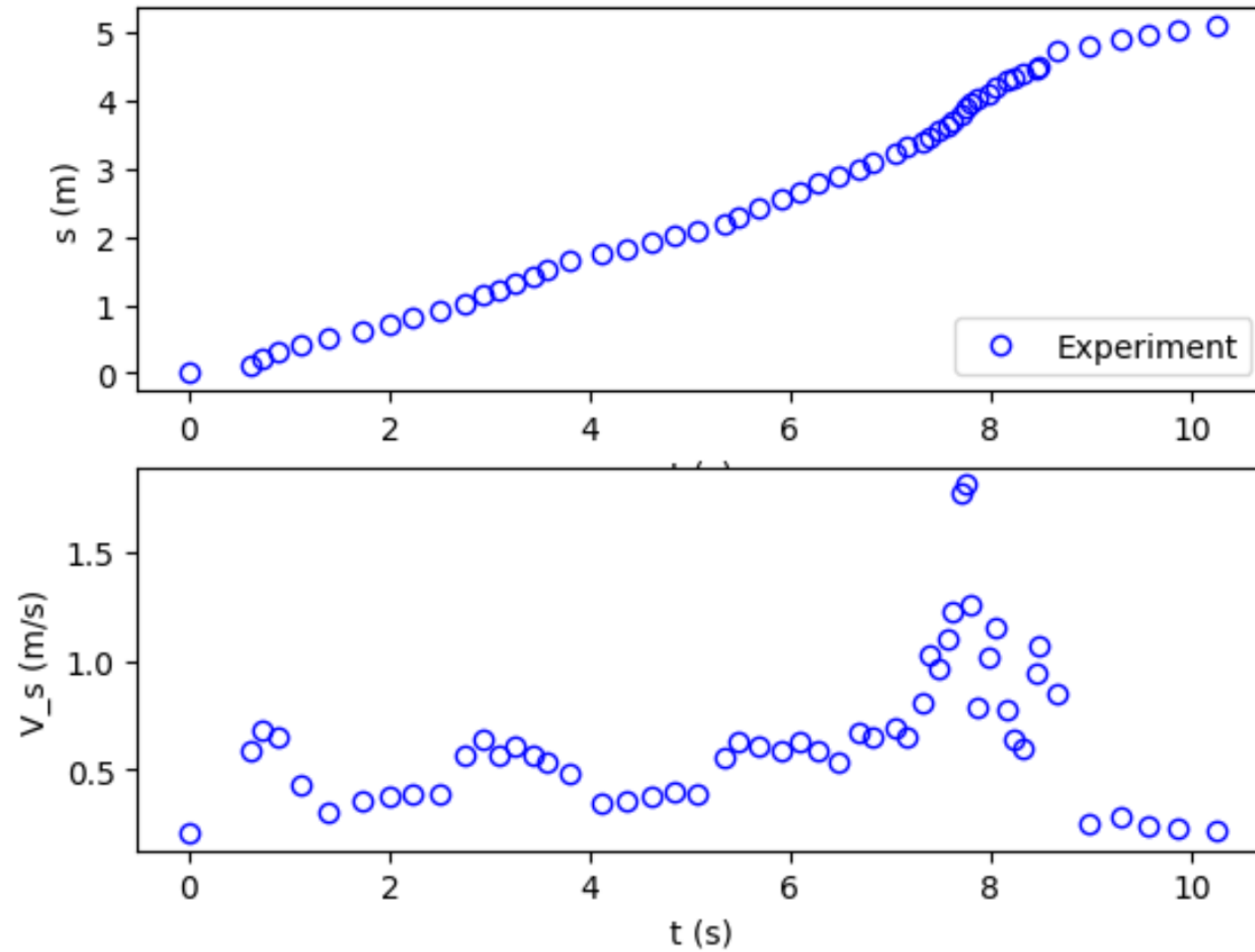


YZ thrilling



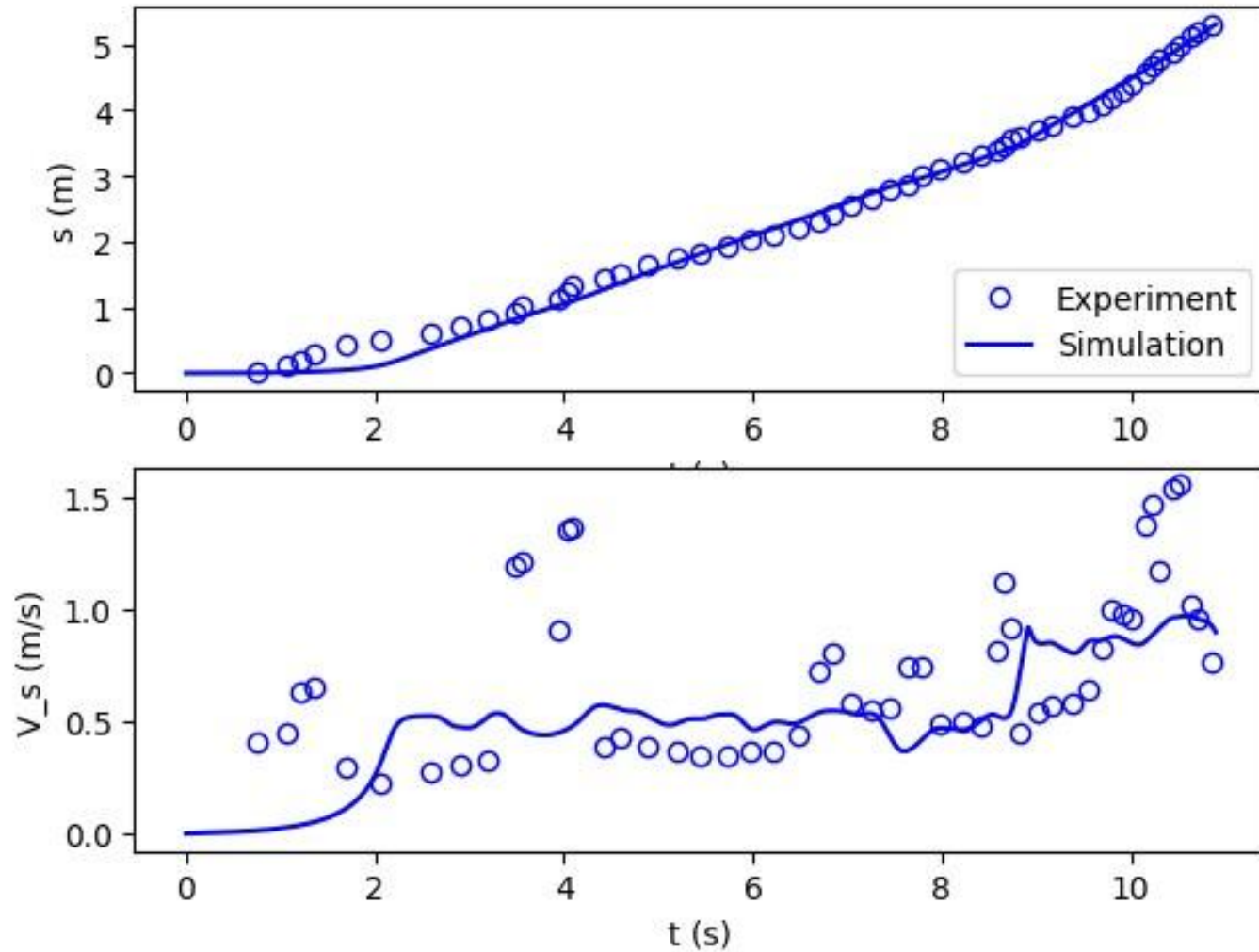


track motion measurements



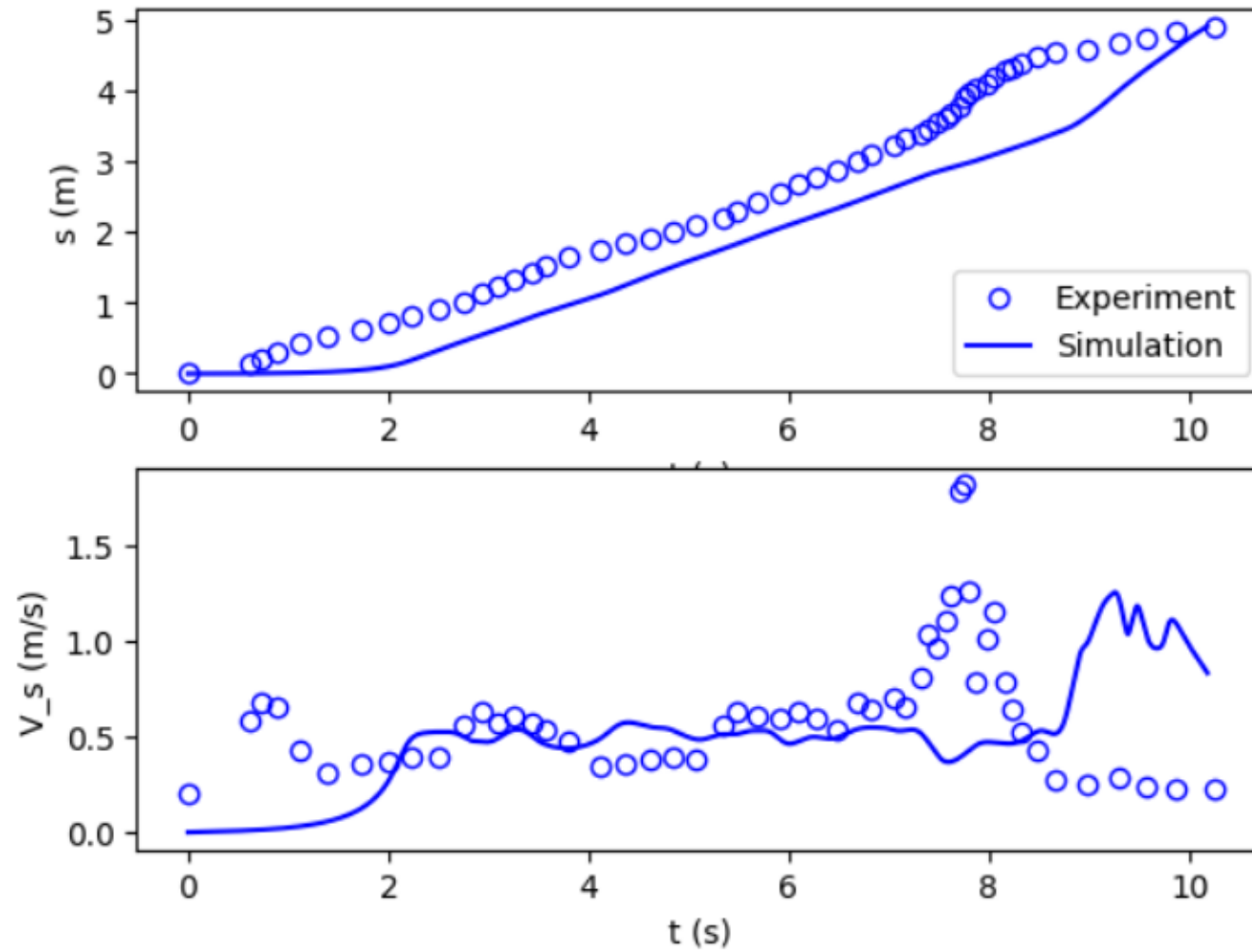


comparison





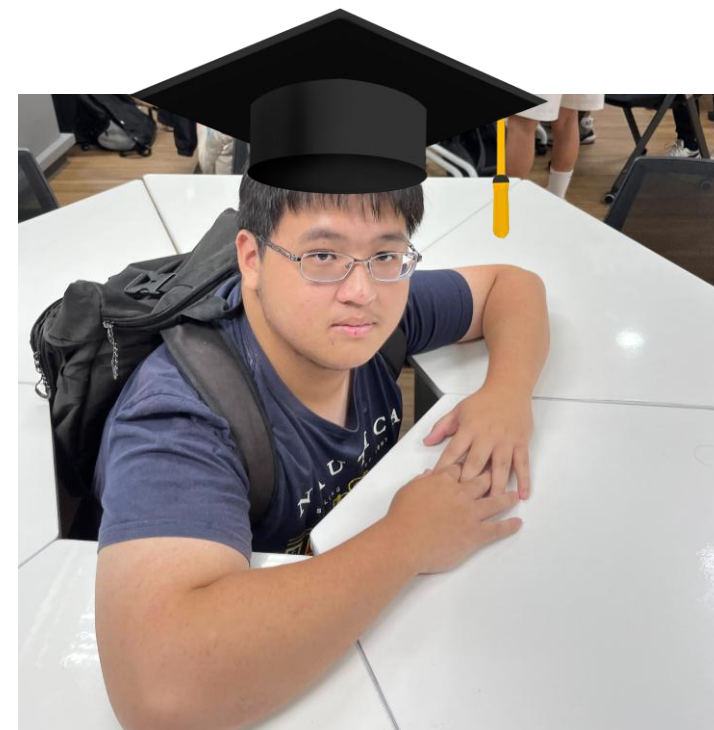
Comparison thrilling



 coding

✈ code

$$\Rightarrow a = \frac{-mg \sin(\beta) - \mu v^2}{m + \frac{\frac{7}{5}mr^2 - \frac{m\ell^2}{4}}{r^2 - \frac{\ell^2}{4}}} - \underbrace{\mu_k g \cos \beta}_{\text{動摩擦係數}}$$



```
cosBeta = (TxTyTz[0]**2+TxTyTz[1]**2)**(1/2)/((TxTyTz[0]**2+TxTyTz[1]**2+TxTyTz[2]**2)**(1/2))
sinBeta = TxTyTz[2]/((TxTyTz[0]**2+TxTyTz[1]**2+TxTyTz[2]**2)**(1/2))
# Calculate gravity component in tangent direction:
gs = -g*TxTyTz[2];

# Sphere acceleration parallel to the track:

# replace by your own equation(s) here !
As = (-ma*g*sinBeta-(0.0159487*(VsSim[k]**2)))/(ma+((2/5)*((ma*(r**2)))/(r**2-((b**2)/4))))-mu * g * cosBeta
#0.0159487 空氣阻力係數
```



comparison

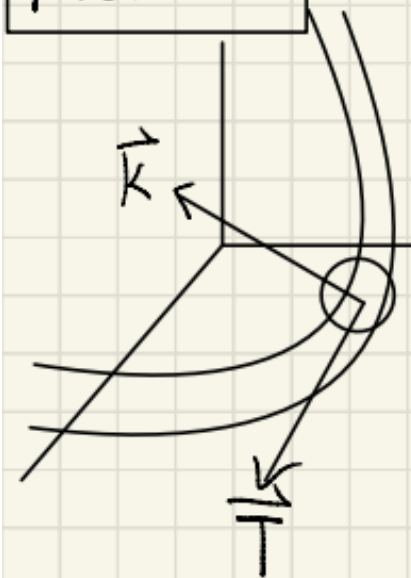
```
xWaypoints = dataWaypoints[:,1][initialpoint:]  
yWaypoints = dataWaypoints[:,2][initialpoint:]  
zWaypoints = dataWaypoints[:,3][initialpoint:]  
bWaypoints = dataWaypoints[:,4][initialpoint:]  
ThetadWaypoints = dataWaypoints[:,5][initialpoint:]
```

請輸入初始點:

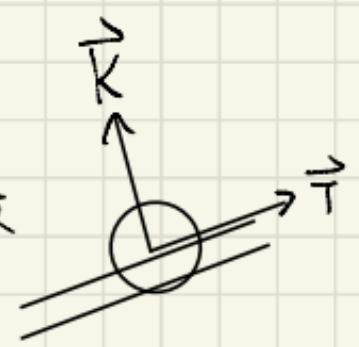
```
TxTyTz = np.array([np.interp(sSim[k], sTrack, Td) for Td in TxTyTzTrack])  
# Interpolate curvature vector at current location:  
KxKyKz = np.array([np.interp(sSim[k], sTrack, Kd) for Kd in KxKyKzTrack])  
cosBeta = (TxTyTz[0]**2+TxTyTz[1]**2)**(1/2)/((TxTyTz[0]**2+TxTyTz[1]**2+TxTyTz[2]**2)**(1/2))  
sinBeta = TxTyTz[2]/((TxTyTz[0]**2+TxTyTz[1]**2+TxTyTz[2]**2)**(1/2))  
# Calculate gravity component in tangent direction:  
gs = -g*TxTyTz[2]  
  
# Sphere acceleration parallel to the track:  
  
# replace by your own equation(s) here !  
As = (-ma*g*sinBeta-(0.0129487*(VsSim[k]**2)))/(ma+(((2/5)*((ma*(r**2))+ma*(r**2-(bSim/2)**2))/(r**2-((bSim**2)/4))))-cosBeta*mu*g
```

 derailment

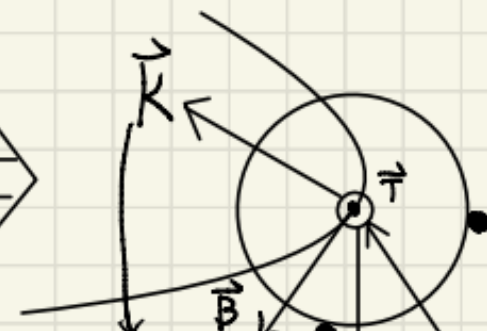
情況 1



截一小段
 ds



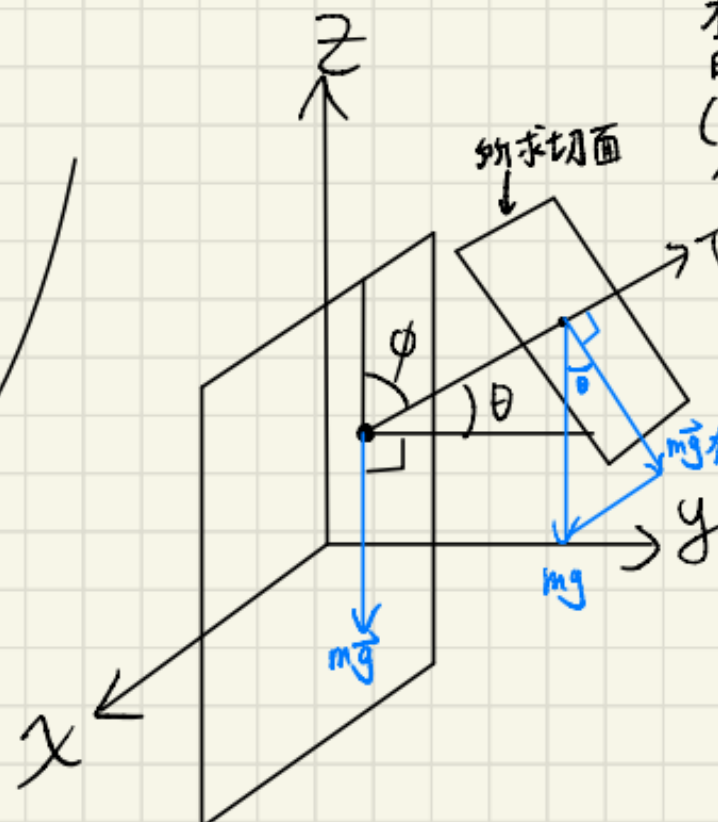
取和 $\vec{T}$ (速度方向) 上的切面



將軌道
看做是一條線
時的曲率向量
(和軌道水平
傾角無關)

$\vec{C}_k$ ($\vec{C}_k$ 和切面上不考慮)

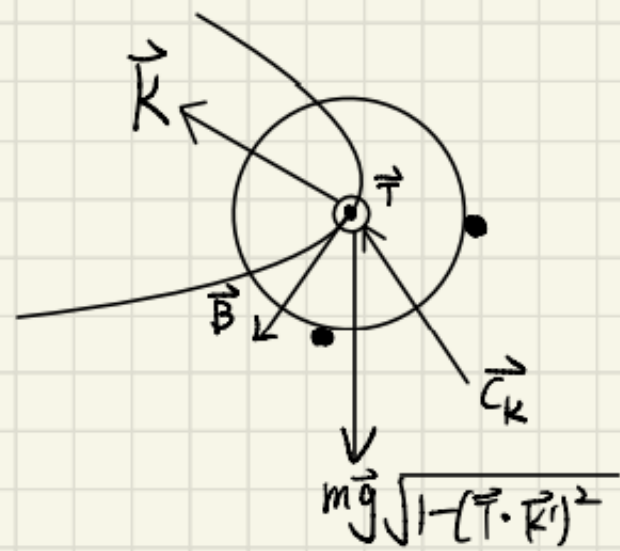
$m\vec{g}$ 在切面上的分量



$$\vec{T} \cdot \vec{k}' = \cos \phi = \sin \theta$$

$$\Rightarrow \vec{k}' = (0, 0, 1)$$

$$m\vec{g} \cos \theta = m\vec{g} \sqrt{1 - (\vec{T} \cdot \vec{k})^2}$$

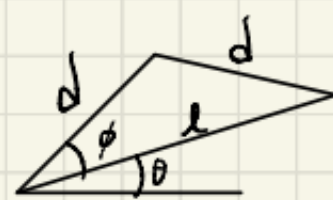


$$\sum \vec{F} = \vec{c}_k + m\vec{g} \sqrt{1 - (\vec{r} \cdot \vec{k})^2} = m \vec{a}$$

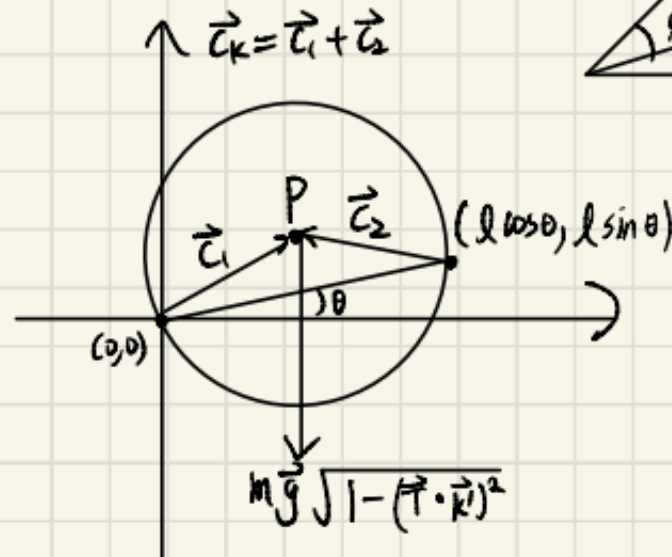
$$\vec{a} = \vec{a}_t + \vec{a}_n = \vec{a}_n = \frac{v^2}{r} \frac{\vec{k}}{|\vec{k}|} = v^2 \vec{k} \quad (|\vec{k}| = \frac{1}{r})$$

切面方向不考慮

分析 $\sum \vec{F}$



$$\cos \phi = \left(\frac{\frac{l}{2}}{d} \right) \Rightarrow \phi = \cos^{-1} \left(\frac{l}{2d} \right)$$



圓心 P 座標 $(d \cos(\theta + \cos^{-1}(\frac{l}{2d})), d \sin(\theta + \cos^{-1}(\frac{l}{2d})))$

$$|\vec{c}_1| = l_1 \quad |\vec{c}_2| = l_2$$

$$\vec{c}_1 = (l_1 \cos(\theta + \cos^{-1}(\frac{l}{2d})), l_1 \sin(\theta + \cos^{-1}(\frac{l}{2d})))$$

$$\vec{c}_2 \text{ 方向 (單位)} = \left(\frac{d \cos(\theta + \cos^{-1}(\frac{l}{2d})) - l \cos \theta}{d}, \frac{d \sin(\theta + \cos^{-1}(\frac{l}{2d})) - l \sin \theta}{d} \right)$$

車寬 l 球半徑 d
 $0 \leq \theta \leq \frac{\pi}{2} \quad l \leq 2d$

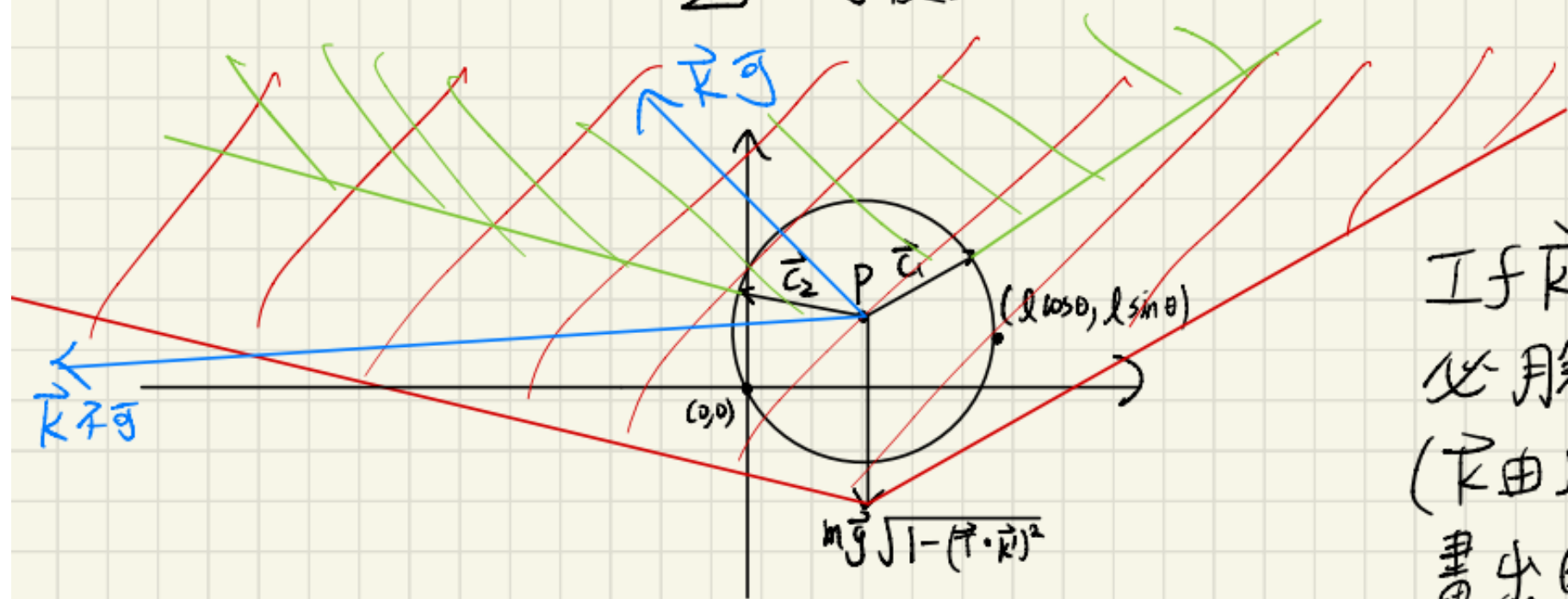
$$\vec{c}_2 = \left(c_2 \cos\left(\theta + \cos^{-1}\left(\frac{d}{2a}\right)\right) - \frac{c_2 l \cos\theta}{d}, c_2 \sin\left(\theta + \cos^{-1}\left(\frac{d}{2a}\right)\right) - \frac{c_2 l \sin\theta}{d} \right)$$

$$m\vec{g}\sqrt{1-(\vec{r}\cdot\vec{k})^2} = (0, -mg\sqrt{1-(\vec{r}\cdot\vec{k})^2})$$

設軌道無敵堅固 $\Rightarrow 0 \leq |\vec{c}_1|, |\vec{c}_2| \leq \infty$

且 $m\vec{g}\sqrt{1-(\vec{r}\cdot\vec{k})^2}$ 為定值

Σ 可覆蓋最大範圍為 



若 $\vec{r}$ 超出 

必脫軌!

($\vec{r}$ 由以一條線畫出的軌道決定)

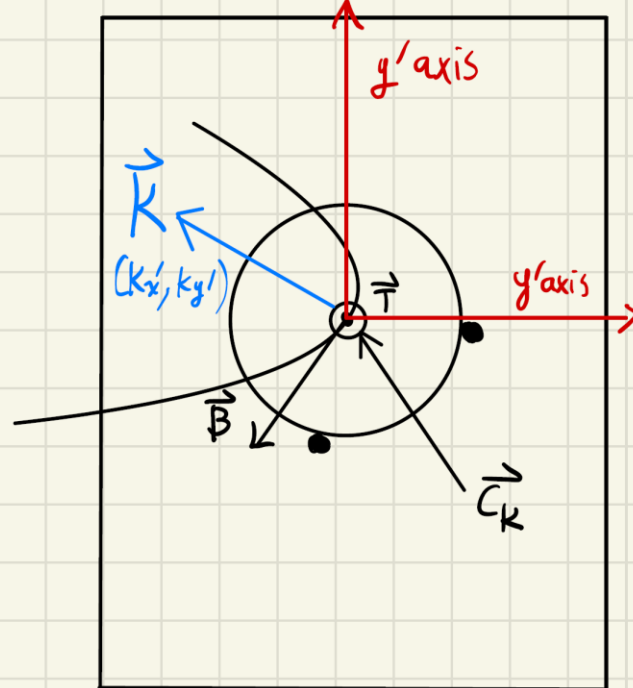
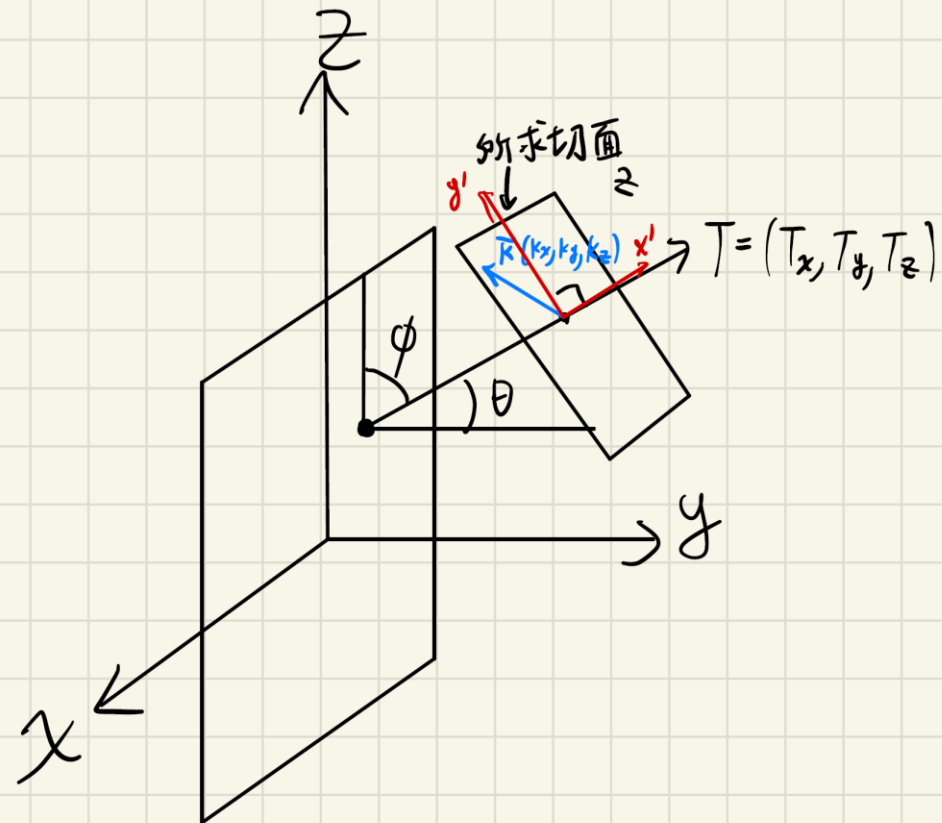
but 原假設又正向力 $\vec{C}_k$ 只有一變數, 在同一條件下必可解出唯一 $\vec{C}_k$

$\Rightarrow \vec{C}_1 + \vec{C}_2 = \vec{C}_k$ C_1, C_2 有 2 個自由度 $\Rightarrow$ 在  中並不是所有 $\vec{C}$ 皆可滿足

$\Rightarrow$ 解 C_1, C_2 之關係

$$\begin{aligned}\sum \vec{F} &= \left((C_1 + C_2) \cos\left(\theta + \cos^{-1}\left(\frac{l}{2d}\right)\right) - \frac{C_2 l \cos\theta}{d}, (C_1 + C_2) \sin\left(\theta + \cos^{-1}\left(\frac{l}{2d}\right)\right) - \frac{C_2 l \sin\theta}{d} - mg\sqrt{1 - T_z^2} \right) \\ &= \left((C_1 + C_2) \left(\frac{l}{2d} \cos\theta - \sqrt{1 - \frac{l^2}{4d^2}} \sin\theta \right) - \frac{C_2 l \cos\theta}{d}, (C_1 + C_2) \left(\frac{l}{2d} \sin\theta + \sqrt{1 - \frac{l^2}{4d^2}} \cos\theta \right) - \frac{C_2 l \sin\theta}{d} - mg\sqrt{1 - T_z^2} \right) \\ &= m v^2 \vec{k}\end{aligned}$$

電月箇可算出 $\vec{k} = (k_x, k_y, k_z) \Rightarrow$ 算 $\vec{k}$ 在垂直 $\vec{T}$ 的平面上的 x, y 分量 (k_x, k_y)
(俯視 $\vec{T}$ 向上為正, 向右為 x 正)



求 y' 方向 条件 ① $\vec{y}' \perp \vec{T}$
 ② $\vec{y}', \vec{T}, \vec{z}$ 共面

$$y'_x T_x + y'_y T_y + y'_z T_z = 0$$

$$T_y y'_x - T_x y'_y = 0$$



面为 $(\vec{T} \times \vec{z}) \cdot \vec{x} = 0$
 $T_y x - T_x y = 0$

$\Rightarrow$ 取其中一组解 $\vec{y}' = (-T_z, -T_z, T_x + T_y)$ * let $y'_z > 0$
向上为正

$$\vec{x}' = \vec{y}' \times \vec{T} = (-T_x T_y - T_y^2 - T_z^2, T_x^2 + T_x T_y + T_z^2, T_x T_z - T_y T_z)$$

$K_{x'}$ 为 $\vec{K}$ 在 $\vec{x}'$ 上的投影 $= \frac{\vec{K} \cdot \vec{x}'}{|\vec{x}'|^2} \vec{x}'$

$\vec{y}'$ 同理 $= \frac{\vec{K} \cdot \vec{y}'}{|\vec{y}'|^2} \vec{y}'$

$$\text{In[9]:= } K_{xxv} = \frac{K \cdot Xp}{\text{Norm}[Xp]}$$

$$\text{Out[9]= } \frac{Kz (Tx Tz - Ty Tz) + Kx (-Tx Ty - Ty^2 - Tz^2) + Ky (Tx^2 + Tx Ty + Tz^2)}{\sqrt{\text{Abs}[Tx Tz - Ty Tz]^2 + \text{Abs}[-Tx Ty - Ty^2 - Tz^2]^2 + \text{Abs}[Tx^2 + Tx Ty + Tz^2]^2}} \Rightarrow \vec{K_x}$$

$$\text{In[11]:= } K_{yyv} = \frac{K \cdot Yp}{\text{Norm}[Yp]}$$

$$\text{Out[11]= } \frac{Kz (Tx + Ty) - Kx Tz - Ky Tz}{\sqrt{\text{Abs}[Tx + Ty]^2 + 2 \text{Abs}[Tz]^2}} \Rightarrow \vec{K_y}$$

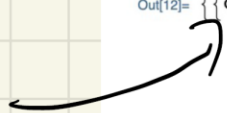
Solve

$$= (c_1 + c_2) \left(\frac{l}{2d} \cos \theta - \sqrt{1 - \frac{l^2}{4d^2}} \sin \theta \right) - \frac{c_2 l \cos \theta}{d}, (c_1 + c_2) \left(\frac{l}{2d} \sin \theta + \sqrt{1 - \frac{l^2}{4d^2}} \cos \theta \right) - \frac{c_2 l \sin \theta}{d} - mg \sqrt{1 - T_z^2}$$

$$= m v^2 \vec{K}$$

脱軌

代表 $c_1 = 0$



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In[12]:= Solve[{{(c1 + c2) (l/(2d) Cos[θ] - Sqrt[1 - l^2/(4d^2)] Sin[θ]) - (c2 l Cos[θ])/d == m v^2 Kxxv, (c1 + c2) (l/(2d) Sin[θ] + Sqrt[1 - l^2/(4d^2)] Cos[θ]) - (c2 l Sin[θ])/d - mg Sqrt[1 - Tz^2] == m v^2 Kyyv}}, {c1, c2}]

Out[12]:= {{c1 -> -((m (Kz (Tx Tz - Ty Tz) + Kx (-Tx Ty - Ty^2 - Tz^2) + Ky (Tx^2 + Tx Ty + Tz^2)) v^2 (Sqrt[1 - l^2/(4d^2)] Cos[θ] - l/(2d) Sin[θ])) / (Sqrt[Abs[Tx Tz - Ty Tz]^2 + Abs[-Tx Ty - Ty^2 - Tz^2]^2 + Abs[Tx^2 + Tx Ty + Tz^2]^2) - (-gm Sqrt[1 - Tz^2] - (m (Kz (Tx + Ty) - Kx Tz - Ky Tz) v^2) / (Sqrt[Abs[Tx + Ty]^2 + 2 Abs[Tz]^2]) (l/(2d) Cos[θ] - Sqrt[1 - l^2/(4d^2)] Sin[θ])) / (l Sqrt[1 - l^2/(4d^2)] Cos[θ]^2 / d + l Sqrt[1 - l^2/(4d^2)] Sin[θ]^2 / d)), c2 -> -((d Ky Sqrt[4d^2 - l^2]/d^2 m Tx^2 v^2 Sqrt[Abs[Tx + Ty]^2 + 2 Abs[Tz]^2] Cos[θ] - d Kx Sqrt[4d^2 - l^2]/d^2 m Tx Ty v^2 Sqrt[Abs[Tx + Ty]^2 + 2 Abs[Tz]^2] Cos[θ] + d Ky Sqrt[4d^2 - l^2]/d^2 m Tx Ty v^2 Sqrt[Abs[Tx + Ty]^2 + 2 Abs[Tz]^2] Cos[θ] - d Kx Sqrt[4d^2 - l^2]/d^2 m Ty^2 v^2 Sqrt[Abs[Tx + Ty]^2 + 2 Abs[Tz]^2] Cos[θ] + d Kz Sqrt[4d^2 - l^2]/d^2 m Tx Tz v^2 Sqrt[Abs[Tx + Ty]^2 + 2 Abs[Tz]^2] Cos[θ] - d Kz Sqrt[4d^2 - l^2]/d^2 m Ty Tz v^2 Sqrt[Abs[Tx + Ty]^2 + 2 Abs[Tz]^2] Cos[θ] - d Kx Sqrt[4d^2 - l^2]/d^2 m Tz^2 v^2 Sqrt[Abs[Tx + Ty]^2 + 2 Abs[Tz]^2] Cos[θ] + d Ky Sqrt[4d^2 - l^2]/d^2 m Tz^2 v^2 Sqrt[Abs[Tx + Ty]^2 + 2 Abs[Tz]^2] Cos[θ] - Kz l m Tx v^2 Sqrt[Abs[Tx Tz - Ty Tz]^2 + Abs[-Tx Ty - Ty^2 - Tz^2]^2 + Abs[Tx^2 + Tx Ty + Tz^2]^2] Cos[θ] - Kz l m Ty v^2 Sqrt[Abs[Tx Tz - Ty Tz]^2 + Abs[-Tx Ty - Ty^2 - Tz^2]^2 + Abs[Tx^2 + Tx Ty + Tz^2]^2] Cos[θ] + Kx l m Tz v^2 Sqrt[Abs[Tx Tz - Ty Tz]^2 + Abs[-Tx Ty - Ty^2 - Tz^2]^2 + Abs[Tx^2 + Tx Ty + Tz^2]^2] Cos[θ] + Ky l m Tz v^2 Sqrt[Abs[Tx Tz - Ty Tz]^2 + Abs[-Tx Ty - Ty^2 - Tz^2]^2 + Abs[Tx^2 + Tx Ty + Tz^2]^2] Cos[θ] - g l m Sqrt[1 - Tz^2] Sqrt[Abs[Tx + Ty]^2 + 2 Abs[Tz]^2] Sqrt[Abs[Tx Tz - Ty Tz]^2 + Abs[-Tx Ty - Ty^2 - Tz^2]^2 + Abs[Tx^2 + Tx Ty + Tz^2]^2] Cos[θ] +

Solve $C_1 = 0$

WOLFRAM MATHEMATICA 學生版

展示 | MathWorld | Wolfram 社群 | 求助

In[15]:= Solve[
求解

$$-\left(-\left(\left(m \left(Kz \left(Tx Tz - Ty Tz\right) + Kx \left(-Tx Ty - Ty^2 - Tz^2\right) + Ky \left(Tx^2 + Tx Ty + Tz^2\right)\right) v^2 \left(\sqrt{1 - \frac{l^2}{4 d^2}} \cos[\theta] - \frac{l \sin[\theta]}{2 d}\right)\right) / \left(\sqrt{\text{Abs}[Tx Tz - Ty Tz]^2 + \text{Abs}[-Tx Ty - Ty^2 - Tz^2]^2 + \text{Abs}[Tx^2 + Tx Ty + Tz^2]^2}\right) - \left(-g m \sqrt{1 - Tz^2} - \frac{m \left(Kz \left(Tx + Ty\right) - Kx Tz - Ky Tz\right) v^2}{\sqrt{\text{Abs}[Tx + Ty]^2 + 2 \text{Abs}[Tz]^2}}\right) \left(-\frac{l \cos[\theta]}{2 d} - \sqrt{1 - \frac{l^2}{4 d^2}} \sin[\theta]\right)\right) / \left(\frac{l \sqrt{1 - \frac{l^2}{4 d^2}} \cos[\theta]^2}{d} + \frac{l \sqrt{1 - \frac{l^2}{4 d^2}} \sin[\theta]^2}{d}\right) = 0,$$

v]

Out[15]=

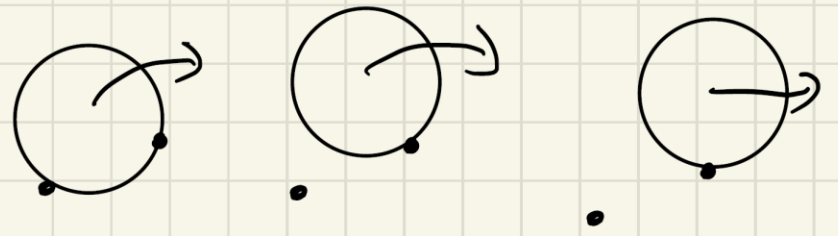
$$\left\{\left\{v \rightarrow -\left(\sqrt{\frac{g l \sqrt{1 - Tz^2} \cos[\theta]}{d} + 2 g \sqrt{1 - \frac{l^2}{4 d^2}} \sqrt{1 - Tz^2} \sin[\theta]}\right) / \left(\sqrt{\left(-\frac{Kz l Tx \cos[\theta]}{d \sqrt{\text{Abs}[Tx + Ty]^2 + 2 \text{Abs}[Tz]^2}} - \frac{Kz l Ty \cos[\theta]}{d \sqrt{\text{Abs}[Tx + Ty]^2 + 2 \text{Abs}[Tz]^2}} + \frac{Kx l Tz \cos[\theta]}{d \sqrt{\text{Abs}[Tx + Ty]^2 + 2 \text{Abs}[Tz]^2}} + \frac{Ky l Tz \cos[\theta]}{d \sqrt{\text{Abs}[Tx + Ty]^2 + 2 \text{Abs}[Tz]^2}} - \frac{2 Ky \sqrt{1 - \frac{l^2}{4 d^2}} Tx^2 \cos[\theta]}{\sqrt{\text{Abs}[Tx Tz - Ty Tz]^2 + \text{Abs}[-Tx Ty - Ty^2 - Tz^2]^2 + \text{Abs}[Tx^2 + Tx Ty + Tz^2]^2}} + \frac{2 Kx \sqrt{1 - \frac{l^2}{4 d^2}} Tx Ty \cos[\theta]}{\sqrt{\text{Abs}[Tx Tz - Ty Tz]^2 + \text{Abs}[-Tx Ty - Ty^2 - Tz^2]^2 + \text{Abs}[Tx^2 + Tx Ty + Tz^2]^2}} - \frac{2 Ky \sqrt{1 - \frac{l^2}{4 d^2}} Tx Ty \cos[\theta]}{\sqrt{\text{Abs}[Tx Tz - Ty Tz]^2 + \text{Abs}[-Tx Ty - Ty^2 - Tz^2]^2 + \text{Abs}[Tx^2 + Tx Ty + Tz^2]^2}}\right)}\right\}$$

Full simplify answer

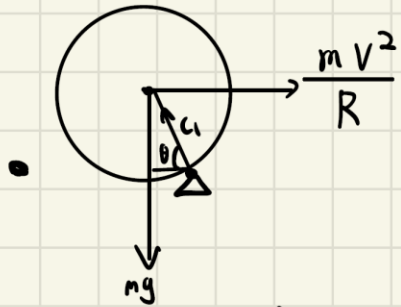
$$\text{Out[14]=} \left(\sqrt{\frac{g \sqrt{1 - Tz^2} \left(l \cos[\theta] + d \sqrt{4 - \frac{l^2}{d^2}} \sin[\theta] \right)}{d}} \right) / \left(\sqrt{\left(\left(l \left(-Kz (Tx + Ty) + (Kx + Ky) Tz \right) \right. \right. \right. \\ \sqrt{\text{Abs}[(Tx - Ty) Tz]^2 + \text{Abs}[Tx (Tx + Ty) + Tz^2]^2 + \text{Abs}[Ty (Tx + Ty) + Tz^2]^2 +} \\ d (Kz (-Tx + Ty) Tz - Ky (Tx (Tx + Ty) + Tz^2) + Kx (Ty (Tx + Ty) + Tz^2)) \\ \left. \left. \sqrt{\frac{(4 d^2 - l^2) (\text{Abs}[Tx + Ty]^2 + 2 Tz \text{Conjugate}[Tz])}{d^2}} \right) \cos[\theta] + \right. \\ \left. \left(d \left(-Kz (Tx + Ty) + (Kx + Ky) Tz \right) \sqrt{\left(\frac{1}{d^2} (4 d^2 - l^2) (\text{Abs}[(Tx - Ty) Tz]^2 + \right. \right. \right. \\ \left. \left. \left. \text{Abs}[Tx (Tx + Ty) + Tz^2]^2 + \text{Abs}[Ty (Tx + Ty) + Tz^2]^2 \right) \right) + \right. \\ \left. l ((Tx + Ty) (Ky Tx - Kx Ty) + Kz (Tx - Ty) Tz + (-Kx + Ky) Tz^2) \right. \\ \left. \left. \sqrt{(Tx + Ty) \text{Conjugate}[Tx + Ty] + 2 Tz \text{Conjugate}[Tz]} \right) \sin[\theta] \right) / \\ \left(d \sqrt{\left((\text{Abs}[(Tx - Ty) Tz]^2 + \text{Abs}[Tx (Tx + Ty) + Tz^2]^2 + \text{Abs}[Ty (Tx + Ty) + Tz^2]^2) \right. \right. \\ \left. \left. (\text{Abs}[Tx + Ty]^2 + 2 Tz \text{Conjugate}[Tz]) \right) \right) \right) \right)$$

when V
大於左邊
算式
⇒開始脫軌

I 若 $C_1 = 0$ 則開始



定球上為參考座標係

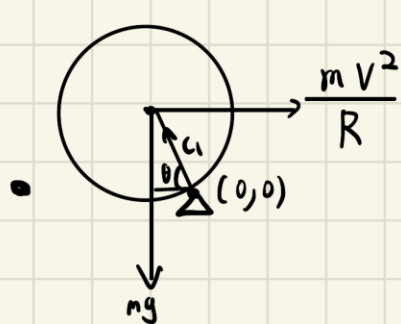


$$\cos^{-1}\left(\frac{l}{2d}\right) \leq \theta \leq \frac{\pi}{2}$$

$$d g m \cos \theta - \frac{d m v^2 \sin \theta}{R} = I \alpha$$

$$I = \frac{2}{5} m d^2 + m d^2 = \frac{7}{5} m d^2$$

$$\alpha = \frac{5(g R \cos \theta - v^2 \sin \theta)}{7 d R}$$



$$\vec{r} = (-d \cos \theta, d \sin \theta, v)$$

$$\sum \vec{F} = \left(\frac{m v^2}{R}, -m g, 0 \right)$$

$$\vec{\tau} = \left(0, 0, d g m \cos \theta - \frac{d m v^2 \sin \theta}{R} \right)$$

$$\omega d\omega = \kappa d\theta$$

$$\frac{d\theta}{dt} = \omega$$

$$\int_0^{\omega} \omega d\omega = \int_{\cos^{-1}(\frac{g}{5d})}^{\frac{\pi}{2}} \frac{5(gR \cos\theta - v^2 \sin\theta)}{7dR} d\theta$$

$$\Rightarrow W(\theta) = \int_{\cos^{-1}(\frac{g}{5d})}^{\frac{\pi}{2}} \frac{d\theta}{\omega(\theta)} = \int_0^t dt$$

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$$\text{In}[] := \int \frac{5 (g * R * \text{Cos}[\theta] - v^2 \text{Sin}[\theta])}{7 d R} d\theta$$

$$\text{Out}[] := \frac{5 v^2 \text{Cos}[\theta]}{7 d R} + \frac{5 g \text{Sin}[\theta]}{7 d}$$

$$\text{In}[] := \frac{5 v^2 \text{Cos}[\theta]}{7 d R} + \frac{5 g \text{Sin}[\theta]}{7 d} // \text{FullSimplify}$$

[完全簡化]

$$\text{Out}[] := \frac{5 (v^2 \text{Cos}[\theta] + g R \text{Sin}[\theta])}{7 d R}$$

$$\text{In}[] := \text{Solve}\left[\left(\frac{5 (v^2 \text{Cos}[\theta] + g R \text{Sin}[\theta])}{7 d R} == \frac{\omega^2}{2}\right), \omega\right]$$

[求解]

$$\text{Out}[] := \left\{\left\{\omega \rightarrow -\frac{\sqrt{\frac{10}{7}} \sqrt{v^2 \text{Cos}[\theta] + g R \text{Sin}[\theta]}}{\sqrt{d} \sqrt{R}}\right\}, \left\{\omega \rightarrow \frac{\sqrt{\frac{10}{7}} \sqrt{v^2 \text{Cos}[\theta] + g R \text{Sin}[\theta]}}{\sqrt{d} \sqrt{R}}\right\}\right\}$$

$$\text{In}[] := -\frac{\sqrt{\frac{10}{7}} \sqrt{v^2 \text{Cos}[\theta] + g R \text{Sin}[\theta]}}{\sqrt{d} \sqrt{R}} // \text{FullSimplify}$$

[完全簡化]

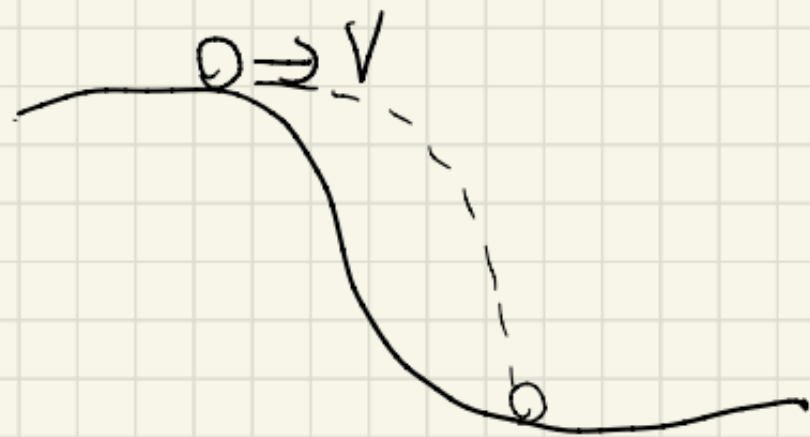
$$\text{Out}[] := -\frac{\sqrt{\frac{10}{7}} \sqrt{v^2 \text{Cos}[\theta] + g R \text{Sin}[\theta]}}{\sqrt{d} \sqrt{R}}$$

V 超過臨界速度後 ($C_1=0$)

$$\begin{aligned} \text{In[*]} &:= \int_{\text{ArcCos}\left[\frac{l}{2d}\right]}^{\frac{\pi}{2}} \frac{1}{\frac{\sqrt{\frac{10}{7}} \sqrt{v^2 \cos[\theta] + g R \sin[\theta]}}{\sqrt{d} \sqrt{R}}} d\theta \\ \text{Out[*]} &= -\sqrt{\frac{7}{10}} \sqrt{d} \sqrt{R} \\ &\quad \left(-\frac{2 \sqrt{g R} \sqrt{\frac{v^4}{g^2 R^2 + v^4}} \text{Hypergeometric2F1}\left[\frac{1}{4}, \frac{1}{2}, \frac{5}{4}, \frac{1}{1 + \frac{v^4}{g^2 R^2}}\right]}{v^2} - \frac{1}{g R \sqrt{1 + \frac{v^4}{g^2 R^2}}} \sqrt{2} \sqrt{g \sqrt{4 - \frac{l^2}{d^2}} R + \frac{l v^2}{d}} \sqrt{\cos\left[\text{ArcCos}\left[\frac{l}{2d}\right] + \text{ArcCot}\left[\frac{g R}{v^2}\right]\right]^2} \right. \\ &\quad \left. \text{Hypergeometric2F1}\left[\frac{1}{4}, \frac{1}{2}, \frac{5}{4}, \sin\left[\text{ArcCos}\left[\frac{l}{2d}\right] + \text{ArcCot}\left[\frac{g R}{v^2}\right]\right]^2\right] \sec\left[\text{ArcCos}\left[\frac{l}{2d}\right] + \text{ArcCot}\left[\frac{g R}{v^2}\right]\right] \right) \text{ if } \text{condition} \end{aligned}$$

↓
經過這一段時間後 ⇒ 真正脫軌! . ⊙ →
(前提是 V 維持不變)

情況 2



$$\text{If } \frac{\vec{V}}{|\vec{V}|} \cdot \vec{K}'_{(0,0,1)} > \vec{T} \cdot \vec{K} \Rightarrow \text{則脫軌} \quad \times$$

$$\Downarrow \cos \theta_1 \quad \Downarrow \cos \theta_2$$

$$\vec{K}'$$

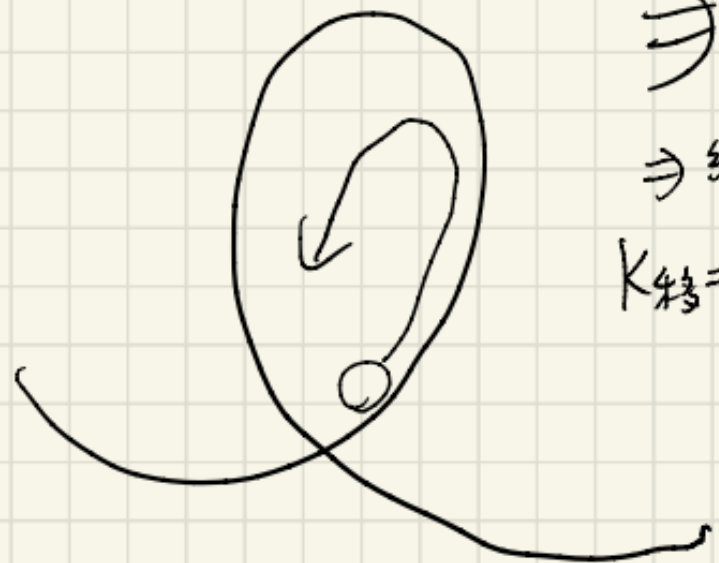
$$\theta_1 \quad \theta_2$$

$$\vec{V}$$

$$\vec{T}$$

單位切向量

情況3



⇒ 設為純滾動 ⇒ 總力學能不變

⇒ 純滾動時 靜摩擦力 不作功 ⇒ 其作用為轉換移動及轉動動能

$$K_{\text{移}} = -fs \Rightarrow K_{\text{轉}} = \tau\theta = fr\theta = fs$$

⇒ 若環頂超出球高 ⇒ 脫軌



contributors and their responsibilities



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slide design



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calculating



薛羽庭
video analysis



陳識凱
slide design



自控

```

def display() -> str:
    global yo_mei_yo_qiu
    lightvalue = light_measure.read_u16()
    value = ball_measure.value()
    if yo_mei_yo_qiu == "the ball is not at the detectable area/":
        color = "yellow"
        ball = "-"
        track = "-"
        yo_mei_yo_qiu = "the ball is at the detectable area/" if lightvalue < 20000 else "the ball is not at the detectable area/"
    elif value == 1 :
        color = "white"
        ball = "ball type: non-metal/"
        track = " track version: mild"
    elif value == 0 :
        color = "white"
        ball = "ball type: metal/"
        track = " track version: thrilling"
    else:
        value = ball_measure.value()
    if yo_mei_yo_qiu == "the ball is at the detectable area/":
        "the ball is at the detectable area/"
    return value, color, ball , track, yo_mei_yo_qiu

```

```

i = 0
yo_mei_yo_qiu = "the ball is not at the detectable area/"
def display() -> str:
    global yo_mei_yo_qiu
    lightvalue = light_measure.read_u16()
    value = ball_measure.value()
    if yo_mei_yo_qiu == "the ball is not at the detectable area/":
        color = "yellow"
        ball = "-"
        track = "-"
        yo_mei_yo_qiu = "the ball is at the detectable area/" if lightvalue < 20000 else "the ball is not at the detectable area/"
    elif value == 1 :
        color = "white"
        ball = "ball type: non-metal/"
        track = "  track version: mild"
    elif value == 0 :
        color = "white"
        ball = "ball type: metal/"
        track = "  track version: thrilling"
    else:
        value = ball_measure.value()
    if yo_mei_yo_qiu == "the ball is at the detectable area/":
        "the ball is at the detectable area/"
    return value, color, ball , track, yo_mei_yo_qiu
#####

```

```

async def echo(request, ws):
    global i # Declare 'i' as a global variable
    while True:
        time.sleep(0.01)
        data, color, ball, track ,yo_mei_yo_qiu = display()
        update_str = f'{data},{color},{ball},{track},{yo_mei_yo_qiu}'
        await ws.send(update_str)
        i = i + 1
        if data == 0:
            for i in range(0, 65535, 100):
                dir_a.duty_u16(65535)
                dir_b.duty_u16(0)
                time.sleep_ms(2)
            dir_a.duty_u16(0)
            for i in range(0, 65535, 100):
                dir_b.duty_u16(65535)
                dir_a.duty_u16(0)
                time.sleep_ms(2)
            dir_b.duty_u16(0)
        elif i == 300: #time limit
            for i in range(0, 65535, 100):
                dir_b.duty_u16(65535)
                dir_a.duty_u16(0)
                time.sleep_ms(2)
            dir_b.duty_u16(0)

```

Thanks for listening

P R E S E N T A T I O N